Xianyu Zhu

Website: zhuxy-cs.github.io/ **Email**: zhuxy@mail.ustc.edu.cn

Research Interests: Machine Learning Systems; Communication-Efficient Distributed Training; Performance Modeling and Optimization; Efficient Multimodal Inference

Technical Skills: C/C++, Python, PyTorch, Linux, CUDA, MPI, OpenMP, NCCL, Verilog

Education

University of Science and Technology of China (C9 League, 985 & 211 University)

- **Master of Computer Science and Technology**, Sep 2023–Jun 2026
- **GPA: 3.87/4.3 (90/100)**
- **Research Group**: Advanced Computer Systems Architecture Laboratory, led by **Prof. Hong An**
- **Selected Coursework**: Foundations of Computer Systems; Parallel Programming; Advanced Computer Networking; Advanced Computer Architecture;
- **Honors and Awards**: First-Class Academic Scholarship, Outstanding Graduates

Northwest A&F University (985 & 211 University)

- **Bachelor of Computer Science and Technology**, Aug 2019–Jun 2023
- **GPA: 3.77/4.0 (90.5/100)** **Rank: 2/132**
- **Research Group**: Parallel and Visual Processing Laboratory, led by **Prof. Bin Liu**
- **Selected Coursework**: Data Structures; Operating Systems; Computer Organization and Architecture; Machine Learning; Cloud Computing;
- **Honors and Awards**: **National Scholarship**; President Scholarship; Outstanding Graduates;

Publications

1. **Xianyu Zhu**, Ruohan Wu, Junshi Chen, Hong An, "Accelerating Large-Scale Distributed Training via Chunk-Wise Gradient Sparsification and Pipelined Communication," Journal of Computer Science and Technology **(JCST)**, 2026.
2. **Xianyu Zhu**, Ruohan Wu, Junshi Chen, Hong An, "swPredictor: A Data-Driven Performance Model for Distributed Data Parallelism Training on Large-Scale HPC Clusters," Performance Evaluation **(PEVA)**, 2025.
3. **Xianyu Zhu**, Jinjiang Li, Runchang Jia, Bin Liu, Zhuohan Yao, Aihong Yuan, Yingqiu Huo, Haixi Zhang, "LAD-Net: A Novel Light Weight Model for Early Apple Leaf Pests and Diseases Classification," IEEE/ACM Transactions on Computational Biology and Bioinformatics **(TCBB)**, 2023.
4. Ruohan Wu, **Xianyu Zhu**, Junshi Chen, Hong An, "SwFormer: Enabling Faster Foundation Models on New Sunway Supercomputer via Holistic Kernel Tiling and Scheduling," Journal of Computer Science and Technology **(JCST)**, 2025.
5. Ruohan Wu, **Xianyu Zhu**, Junshi Chen, Hong An, "SWattention: designing fast and memory-efficient attention for a new Sunway Supercomputer," The Journal of Supercomputing **(TJSC)**, 2024.
6. Jinjiang Li, **Xianyu Zhu**, Runchang Jia, Bin Liu, Cong Yu, "Apple-YOLO: A Novel Mobile Terminal Detector Based on YOLOv5 for Early Apple Leaf Diseases," IEEE International Conference on Computers, Software, and Applications **(COMPSAC)**, 2022.
7. Xinbin Yuan, Cong Yu, Bin Liu, Henan Sun, **Xianyu Zhu**, "CGAN-IRB: a novel data augmentation method for apple leaf diseases," IEEE International Conference on Computers, Software, and Applications **(COMPSAC)**, 2021.

8. Bin Liu, Runchang Jia, **Xianyu Zhu**, Cong Yu, Zhuohan Yao, Haixi Zhang, Dongjian He, "Lightweight identification model for apple leaf diseases and pests based on mobile terminals," Transactions of the Chinese Society of Agricultural Engineering **(TCSAE)**, 2022.

Research Projects

1. **AI-Based Early Diagnosis of Apple Leaf Pests and Diseases Using an Inspection Robot**
   Project Lead | Provincial Innovation and Entrepreneurship Training Program, 2021–2022
   - Propose asymmetric dilated convolution and develop LAD-Inception, a multi-scale feature extraction module.
   - Deploy the model on an NVIDIA Jetson TX2 and integrate it with a wheeled platform to enable autonomous orchard inspection.
   - Build a cloud-edge collaboration workflow for mobile inference, remote monitoring, and result synchronization.
2. **Supernode Training and Inference System for Unified Multimodal Large Models**
   Lead Proposal Writer | Funding Proposal Stage
   - Draft the funding proposal, structure the project's research objectives and work packages, and conduct preliminary research on supernode-based training and inference for unified multimodal large models.
3. **Dataflow Programming and Runtime System Framework Adapted to Domestic Heterogeneous System Architectures**
   Project Member | Strategic Priority Program of Chinese Academy of Sciences
   - Develop a data-driven performance model to accurately predict operator computation and gradient communication overheads in large-scale cluster training.
   - Design a chunk-wise compression algorithm and a pipelined sparse communication framework, achieving 3× to 4.74× communication speedup while reducing peak memory usage.
   - Extend the chunking strategy to PowerSGD and sketch-based algorithms, systematically analyzing their feasibility boundaries and operational overheads across different compression paradigms.
4. **Additional Research Projects in Apple Leaf Disease Detection** | *Project Member*
   - AI-Based Early Monitoring and Warning of Apple Leaf Diseases and Inspection Equipment Development
     Shaanxi Provincial Key Research Program, 2021–2022
   - Apple Leaf Disease Identification and Detection Using Cloud Collaboration and CNNs
     Provincial Innovation and Entrepreneurship Training Program, 2020–2021

Experience & Academic Service

- **Reviewer**, IEEE Transactions on Parallel and Distributed Systems (TPDS) **2026–Present**
- **Research Assistant**, Systems and Networking Lab, The University of Hong Kong, advised by Prof. Heming Cui; worked on distributed training systems for multimodal foundation models **Jul 2025–Jan 2026**
- Participant, Hong Kong PhD Fellowship Summer Workshop (CUHK) **Jul 2025**
- Participant, CCF HPC China 2023 (Qingdao, China) **Aug 2023**
- Teaching Assistant for "Parallel Computing" **Autumn 2022**
- Tedu AI Training Program **Summer 2022**
- Baidu AI Talents Camp **Summer 2021**

Competition

- 11th Student RDMA Programming Competition **Third Prize**
- 2022 Mathematical Contest in Modeling (MCM) **Meritorious Winner**
- 12th Blue Bridge Cup Competition (Provincial Level) **First Prize**
- 2021 Group Programming Ladder Tournament (National Level) **Third Prize**
- 2022 Chinese Collegiate Computing Competition (Northwest Regional Contest) **First Prize**